

Name: Key

4. a) Use row reduction to put the following matrix in reduced row echelon form.

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 10 \\ 5 & 12 & 10 \end{pmatrix} \xrightarrow{R3-5R1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 10 \\ 0 & 2 & 10 \end{pmatrix} \xrightarrow{R3-R2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 10 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{1/2 R2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R1-2R2} \begin{pmatrix} 1 & 0 & -10 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{RREF}$$

b) Write the following linear system over $\mathbb{R}$ as an augmented matrix. Then find the general solution to the system and write it in vector form.

$$\begin{aligned} x + 2y - 3z &= 10 \\ 2x + y - 6z &= 8 \\ 4x + 5y - 12z &= 28 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 10 \\ 2 & 1 & -6 & 8 \\ 4 & 5 & -12 & 28 \end{array} \right) \xrightarrow{R2-2R1} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 10 \\ 0 & -3 & 0 & -12 \\ 4 & 5 & -12 & 28 \end{array} \right)$$

$$\xrightarrow{R3-4R1} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 10 \\ 0 & -3 & 0 & -12 \\ 0 & -3 & 0 & -12 \end{array} \right) \xrightarrow{R3-R2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 10 \\ 0 & -3 & 0 & -12 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{-\frac{1}{3}R2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 10 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\xrightarrow{R1-2R2} \left(\begin{array}{ccc|c} 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightsquigarrow \begin{aligned} x - 3z &= 2 \\ y &= 4 \\ 0 &= 0 \end{aligned} \quad \text{or}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + z \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

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5. a) Compute the inverse of the following real matrices; or explain why an inverse does not exist.

$$A = \begin{pmatrix} 5 & 0 \\ 4 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 0 \\ -4 & 5 \end{pmatrix} = \begin{pmatrix} 1/5 & 0 \\ -4/5 & 1 \end{pmatrix}$$

For B^{-1} :

$$\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 6 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \xrightarrow{R_2 - 6R_1} \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -6 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array} \xrightarrow{-R_3}$$

$$\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -6 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}$$

$$\text{so } B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -6 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

b) Give an example of two matrices A and B such that A is invertible, B is invertible, but $A + B$ is not invertible; or explain why this is not possible.

Many examples exist. For instance:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad A + B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad A + B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad A + B = \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix}$$